

LLM & Generative AI Training – Experimental Summary

1. Introduction

This exercise focused on exploring the practical implementation of Large Language Models (LLMs) using an API-based approach. The objective was to understand how prompt design, parameter tuning, and API interactions influence model behavior and output quality.

2. Experiments Performed

During the exercise, multiple experiments were conducted:

- **Prompt Variations:** Compared unstructured prompts with structured formats ([Context, Task, Constraints, Output]).
 - **Parameter Tuning:** Adjusted temperature and max token values to observe changes in output style and length.
 - **Use Case Testing:** Applied the model to different tasks including summarization, sentiment analysis, and code generation.
 - **Retry Mechanism Testing:** Observed system behavior under simulated API failures such as rate limits and timeouts.
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3. Key Observations

- Structured prompts produced significantly more consistent and relevant outputs compared to unstructured prompts.
 - Lower temperature values resulted in deterministic and factual responses, while higher values increased creativity but reduced reliability.
 - Increasing max tokens increased response length but did not always improve quality.
 - The model showed sensitivity to small prompt changes, affecting output significantly.
 - Retry mechanisms improved system stability under API constraints.
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4. Learnings

- Prompt engineering is critical for controlling LLM output and ensuring consistency.
 - API abstraction simplifies interaction and allows flexibility in model usage.
 - Parameter tuning provides control over response creativity, length, and cost.
 - LLMs are powerful but require validation and structured handling for reliable usage.
 - Building production-ready AI systems requires error handling and resilience strategies.
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5. Challenges Faced

- Encountered rate limiting (429 errors) during repeated API calls.
 - Occasional inconsistent or hallucinated outputs from the model.
 - Difficulty in balancing creativity and accuracy using temperature settings.
 - Debugging API request formatting and response parsing.
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6. Conclusion

This exercise provided hands-on experience in working with LLMs in a practical environment. It highlighted the importance of structured prompting, parameter optimization, and system reliability. These insights are essential for developing scalable and production ready AI applications.